Building the Structure of a LaTeX Document

LaTeX Learning Module

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Abstract

This example introduces the basic parts of a LaTeX document. It demonstrates how to choose a document class, set common options, create a title, write an abstract, and organize information with section headings.

Contents

1 Understanding the Document Class

Every LaTeX document begins by selecting a document class. The class determines the general structure and formatting rules used by the document.

- `article` is useful for short documents, assignments, and reports that do not need chapters.
- `report` is suitable for longer work that can be divided into chapters.
- `book` is designed for book-length projects with front matter and chapters.

The class is selected with the `\documentclass` command:

```
\documentclass[12pt,a4paper]{article}
```

The options in square brackets set the main font size and paper format. Common font-size choices include `10pt`, `11pt`, and `12pt`. A document can also use paper options such as `a4paper` or `letterpaper`.

2 The Preamble and Document Body

A LaTeX source file has two main parts. The preamble appears before `\begin{document}` and contains the document class, package imports, and settings. The body begins after `\begin{document}` and contains the material that will appear in the finished document.

```
\documentclass[11pt,a4paper]{article}
\usepackage[margin=25mm]{geometry}

\begin{document}
Your document content goes here.
\end{document}
```

Commands that load packages or define document-wide settings normally belong in the preamble.

3 Adding a Title and Author Details

The commands `\title`, `\author`, and `\date` store the information that appears in the title area. The command `\maketitle` prints that information.

```
\title{A Short Research Report}
\author{Jordan Lee}
\date{November 2026}
\maketitle
```

If the date field is left empty, use `\date{}` to omit it. The appearance of the title area depends partly on the selected document class.

4 Writing an Abstract

An abstract gives readers a brief overview of the document. In the `article` class, it is usually created with the `abstract` environment.

```
\begin{abstract}
This paper describes the purpose, approach, and main
points of the work in a concise form.
\end{abstract}
```

Keep the abstract focused on the subject and the main information readers should know before reading the full document.

5 Organizing Content with Headings

LaTeX provides commands for arranging material into a clear hierarchy. For an article, common levels include:

- `\section{...}` for a main topic
- `\subsection{...}` for a part of that topic
- `\subsubsection{...}` for a more detailed point

For example:

```
\section{Methods}
\subsection{Materials}
\subsubsection{Equipment}
```

The available heading levels depend on the document class. The `report` and `book` classes also support chapters.

6 A Complete Starter Example

The following short source brings the main ideas together:

```
\documentclass[12pt,a4paper]{article}
\usepackage[margin=25mm]{geometry}

\title{My First Structured Document}
\author{Jordan Lee}
\date{\today}

\begin{document}
\maketitle

\begin{abstract}
A short summary of the document goes here.
```

```
\end{abstract}

\section{Introduction}
Introduce the subject.

\section{Main Discussion}
Explain the important points.

\subsection{A Supporting Detail}
Add a more focused explanation.

\section{Conclusion}
Summarize the main message.
\end{document}
```

7 Key Points

A well-organized LaTeX file starts with a suitable class, places setup commands in the preamble, and keeps the actual content inside the document body. Titles, abstracts, and heading commands help readers navigate the finished work.